Peer Residence
HVAC Load Calculations

for

Elite Software

RHVAC Residential HVAC Loads

Prepared By:

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Saturday, February 25, 2023

Project Report

General Project Information

Project Title:	Peer Residence
Designed By:	Hafiz Muhammad Talha
Project Date:	Wednesday, February 22, 2023
Company Representative:	Hafiz Muhammad Talha
Company City:	Karachi, Pakistan
Company Phone:	+92 310 1002379
Company E-Mail Address:	talhasabir46@gmail.com

Design Data

Reference City:		Louisville, Kentucky
Daily Temperature Range:		Medium
Latitude:	38	Degrees
Elevation:	477	ft.
Altitude Factor:	0.983	
Elevation Sensible Adj. Factor:	1.000	
Elevation Total Adj. Factor:	1.000	
Elevation Heating Adj. Factor:	1.000	
Elevation Heating Adj. Factor:	1.000	

	Outdoor Dry Bulb	Outdoor Wet Bulb	Indoor Rel.Hum	Indoor Dry Bulb	Grains Difference
Winter:	10	0	30	70	26
Summer:	93	74	50	75	32

Check Figures

Total Building Supply CFM:	1,084	CFM Per Square ft.:	4.817
Square ft. of Room Area:	225	Square ft. Per Ton:	38
Volume (ft³) of Cond. Space:	0	Air Turnover Rate (per hour):	0.0

Building Loads

Total Heating Required With Outside Air:	48,418	Btuh	48.418	MBH
Total Sensible Gain:	33,945	Btuh	66	%
Total Latent Gain:	17,840	Btuh	34	%
Total Cooling Required With Outside Air:	51,785	Btuh	4.32	Tons (Based On Sensible + Latent)
			5.95	Tons (Based On 75% Sensible Capacity)

Notes

Calculations are based on 8th edition of ACCA Manual J.
All computed results are estimates as building use and weather may vary.
Be sure to select a unit that meets both sensible and latent loads.

Miscellaneous Report

System 1 System Input Data	Outdoor Dry Bulb	Outdoor Wet Bulb	Indoor Rel.Hum	Indoor Dry Bulb	Grains Difference
Winter:	10	0	30	70	25.64
Summer:	93	74	50	75	32.22

Duct Sizing Inputs

	Main Trunk		Runouts	
Calculate:	Yes		Yes	
Use Schedule:	Yes		Yes	
Roughness Factor:	0.00300		0.01000	
Pressure Drop:	0.1000	in.wg./100 ft.	0.1000	in.wg./100 ft.
Minimum Velocity:	650	ft./min	450	ft./min
Maximum Velocity:	900	ft./min	750	ft./min
Minimum Height:	0	in.	0	in.
Maximum Height:	0	in.	0	in.

Outside Air Data

	Winter		Summer	
Infiltration:	0.000	AC/hr	0.000	AC/hr
Above Grade Volume:	X 0	Cu.ft.	X 0	Cu.ft.
	0	Cu.ft./hr	0	Cu.ft./hr
	X 0.0167		X 0.0167	
Total Building Infiltration:	0	CFM	0	CFM
Total Building Ventilation:	540	CFM	540	CFM

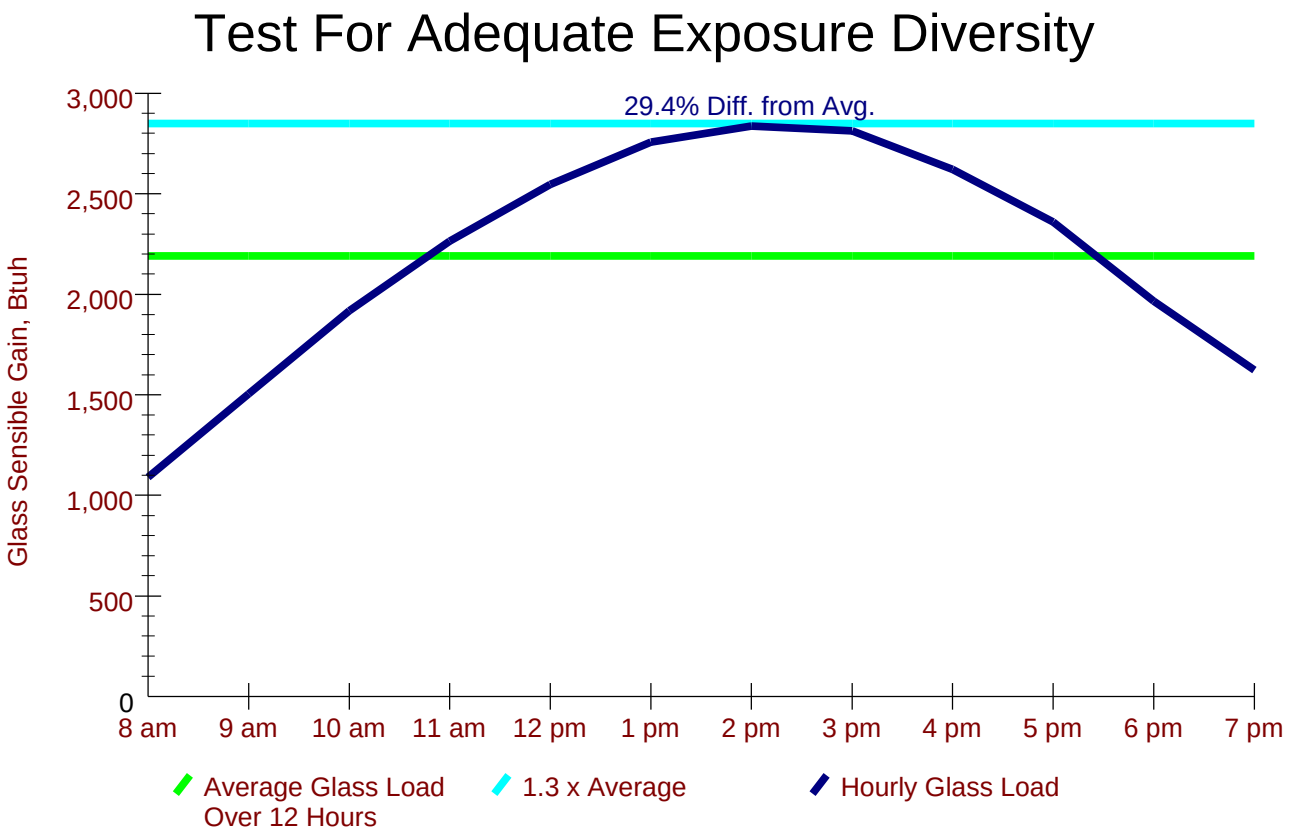
---System 1---

Infiltration & Ventilation Sensible Gain Multiplier:	19.46	=	(1.10 X 0.983 X 18.00 Summer Temp. Difference)
Infiltration & Ventilation Latent Gain Multiplier:	21.54	=	(0.68 X 0.983 X 32.22 Grains Difference)
Infiltration & Ventilation Sensible Loss Multiplier:	64.87	=	(1.10 X 0.983 X 60.00 Winter Temp. Difference)

Load Preview Report

Scope	Area	Sens Gain	Lat Gain	Net Gain	Sens Loss	Win CFM	Sum CFM	Sys CFM	Duct Size
Building: 4.32 Net Tons, 5.95 Recommended Tons, 38 ft.²/Ton, 48.42 MBH Heating									
Building	225	33,945	17,840	51,785	48,418	177	1,084	1,084	
System 1: 4.32 Net Tons, 5.95 Recommended Tons, 38 ft.²/Ton, 48.42 MBH Heating									
System 1	225	33,945	17,840	51,785	48,418	177	1,084	1,084	12x16
Ventilation		10,509	11,630	22,139	35,030				
Zone 1	225	23,436	6,210	29,646	13,388	177	1,084	1,084	
1-Basement	225	9,888	3,220	13,108	10,368	137	457	457	4-6
2-Ground Floor	0	13,548	2,990	16,538	3,020	40	627	627	6-6

System 1 - System - Adequate Exposure Diversity Test



AED Calculation Summary

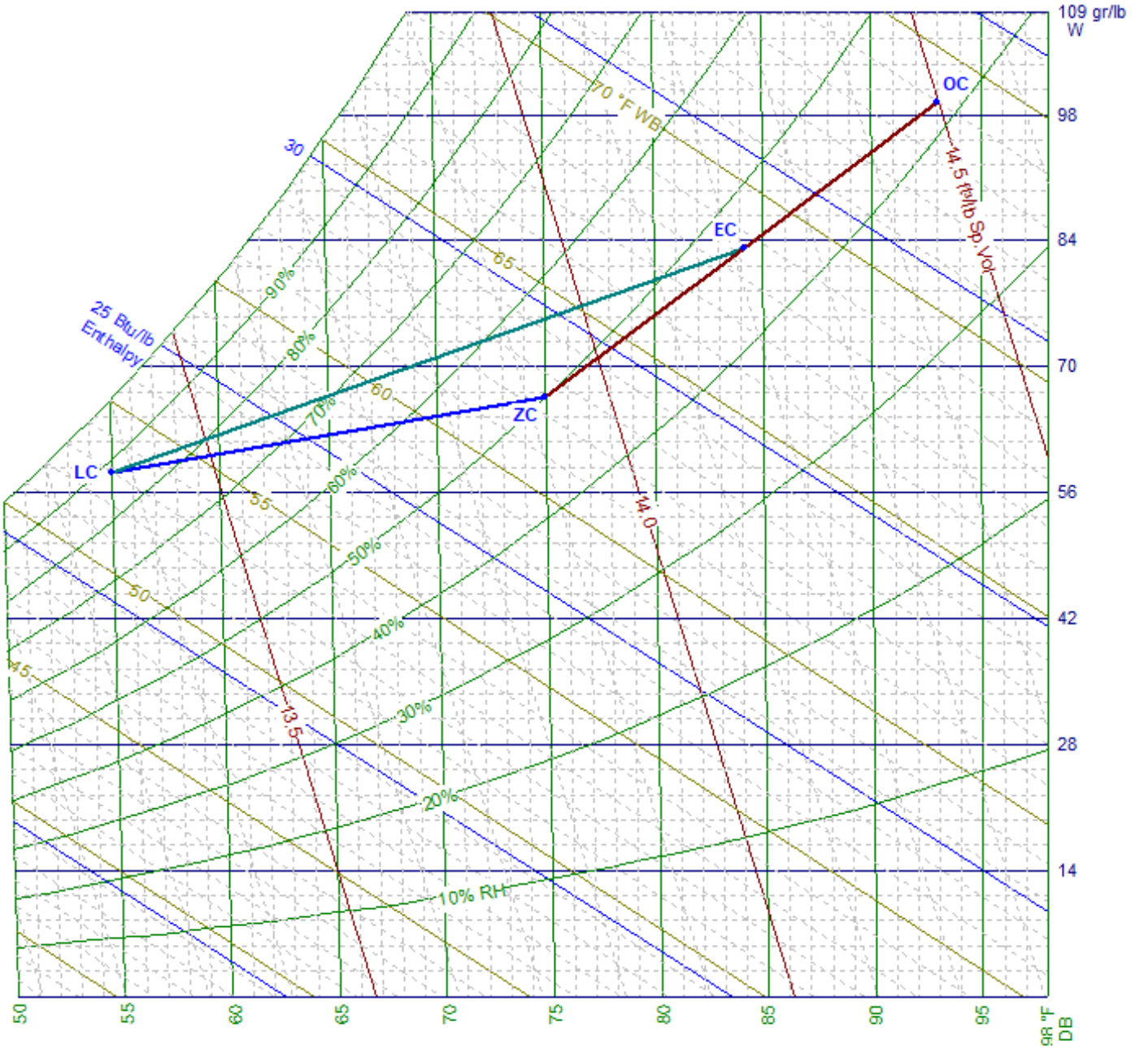
--- SYSTEM HAS ADEQUATE EXPOSURE DIVERSITY. ---

System is on NE, SE, SW, NW rosette.
Peak load exceeds 12-hour average load by 29.4%.
AED Excursion (amount by which peak exceeds 1.3 x average): 0 Btuh

Definition: A system has adequate exposure diversity if the peak-hour glass load for the entire conditioned space does not exceed the average glass load for the entire conditioned space by more than 30 percent.

System 1 - System - Psychrometric Chart

ZC	Zone (Room) Condition	OC	Outdoor Condition
LC	Leaving Coil Condition	EC	Entering Coil Condition
SD	Supply Duct Temperature Rise	DTF	Draw Through Fan Sensible Gain



Total Building Summary Loads

Component Description	Area Quan	Sen Loss	Lat Gain	Sen Gain	Total Gain
SL: Glazing-Skylight, Double Glaze, Low e Glass, horizontal, u-value 0.55	32	1,056	0	2,658	2,658
Roof: Roof/Ceiling-	1723	1,964	0	1,408	1,408
Basment Floor: Floor-	1728	10,368	0	0	0
Subtotals for structure:		13,388	0	4,066	4,066
People:	27		6,210	8,100	14,310
Equipment:			0	0	0
Lighting:	3305			11,270	11,270
Ductwork:		0	0	0	0
Infiltration: Winter CFM: 0, Summer CFM: 0		0	0	0	0
Ventilation: Winter CFM: 540, Summer CFM: 540		35,030	11,630	10,509	22,139
Total Building Load Totals:		48,418	17,840	33,945	51,785

Check Figures

Total Building Supply CFM:	1,084	CFM Per Square ft.:	4.817
Square ft. of Room Area:	225	Square ft. Per Ton:	38
Volume (ft³) of Cond. Space:	0	Air Turnover Rate (per hour):	0.0

Building Loads

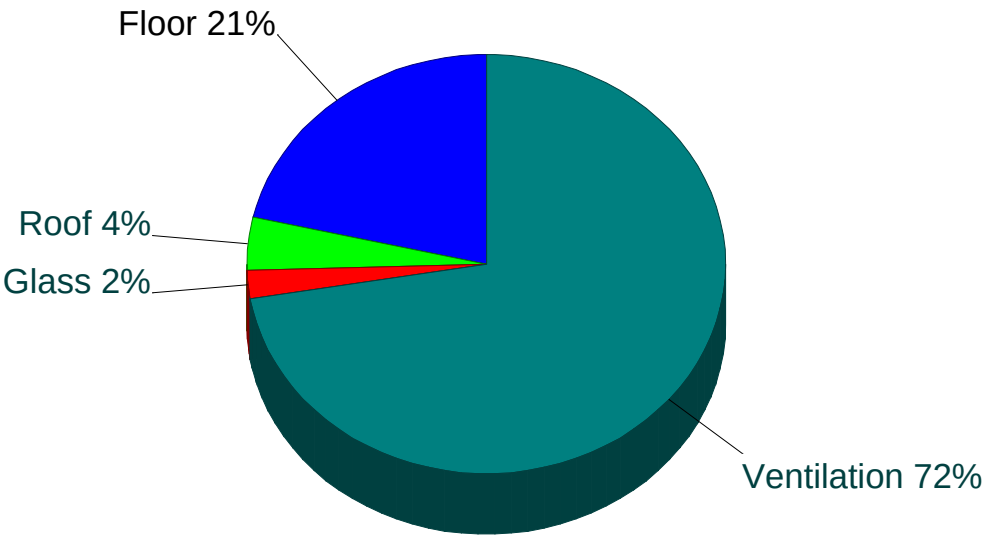
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Total Latent Gain:	17,840	Btuh	34	%
Total Cooling Required With Outside Air:	51,785	Btuh	4.32	Tons (Based On Sensible + Latent)
			5.95	Tons (Based On 75% Sensible Capacity)

Notes

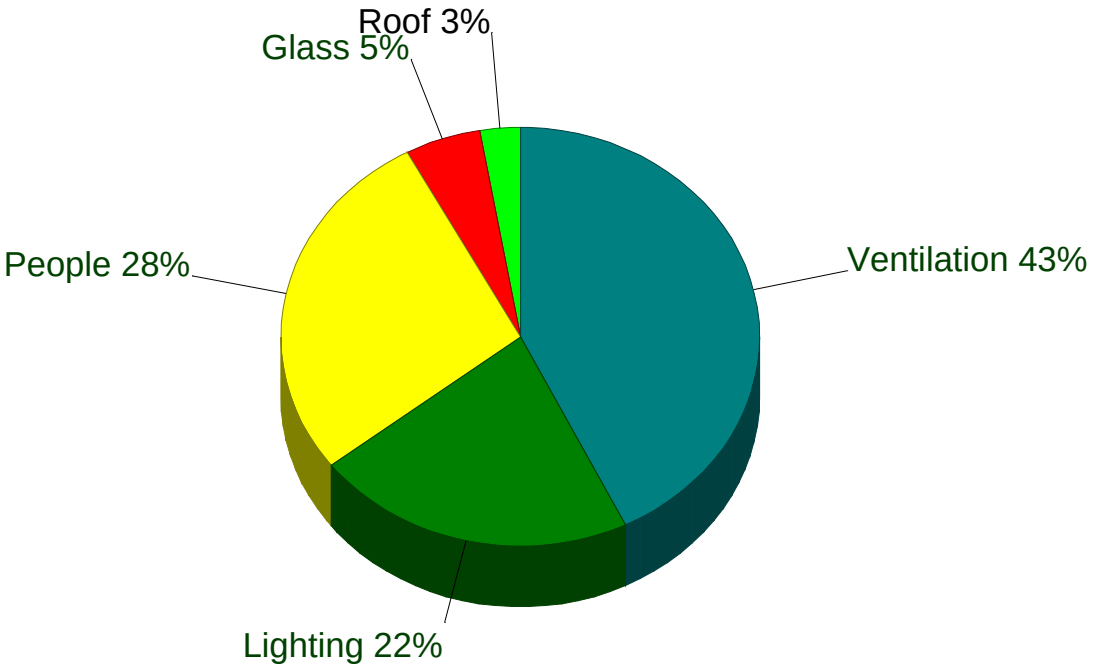
Calculations are based on 8th edition of ACCA Manual J.
All computed results are estimates as building use and weather may vary.
Be sure to select a unit that meets both sensible and latent loads.

Building Pie Chart

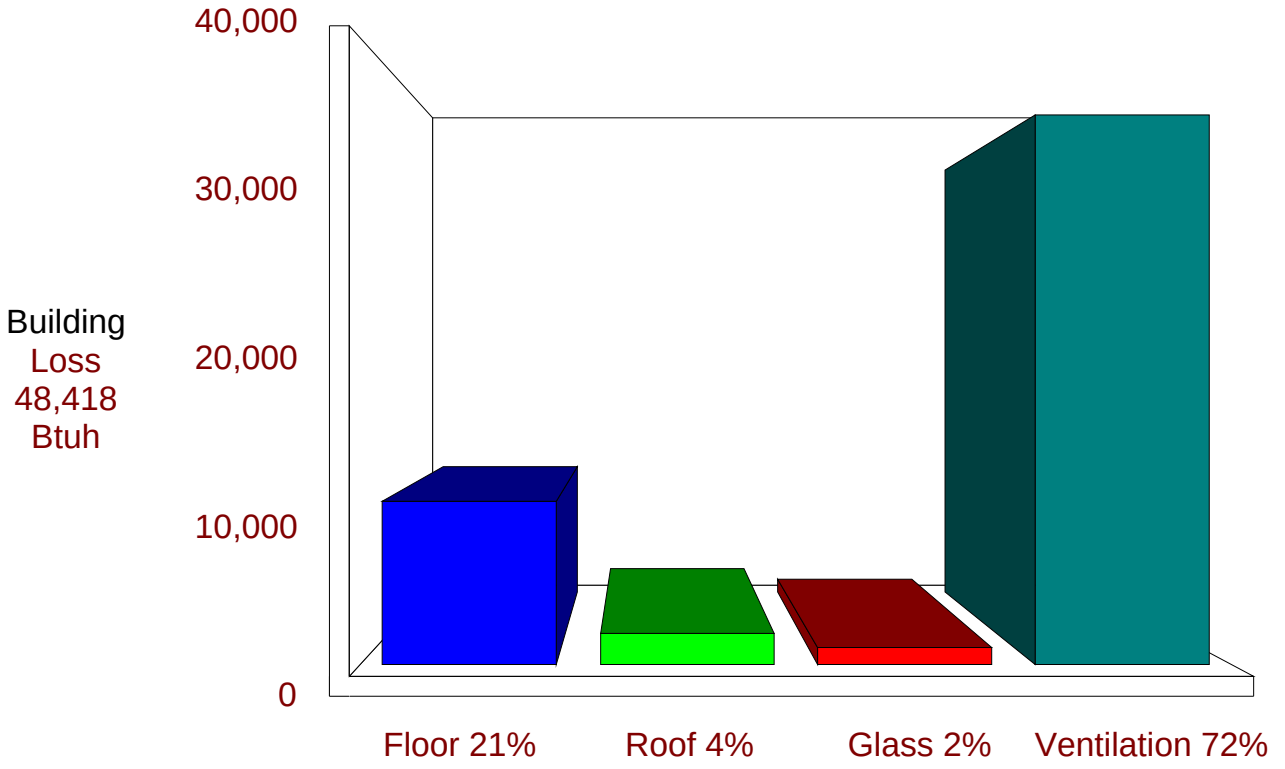
Building
Loss
48,418
Btuh

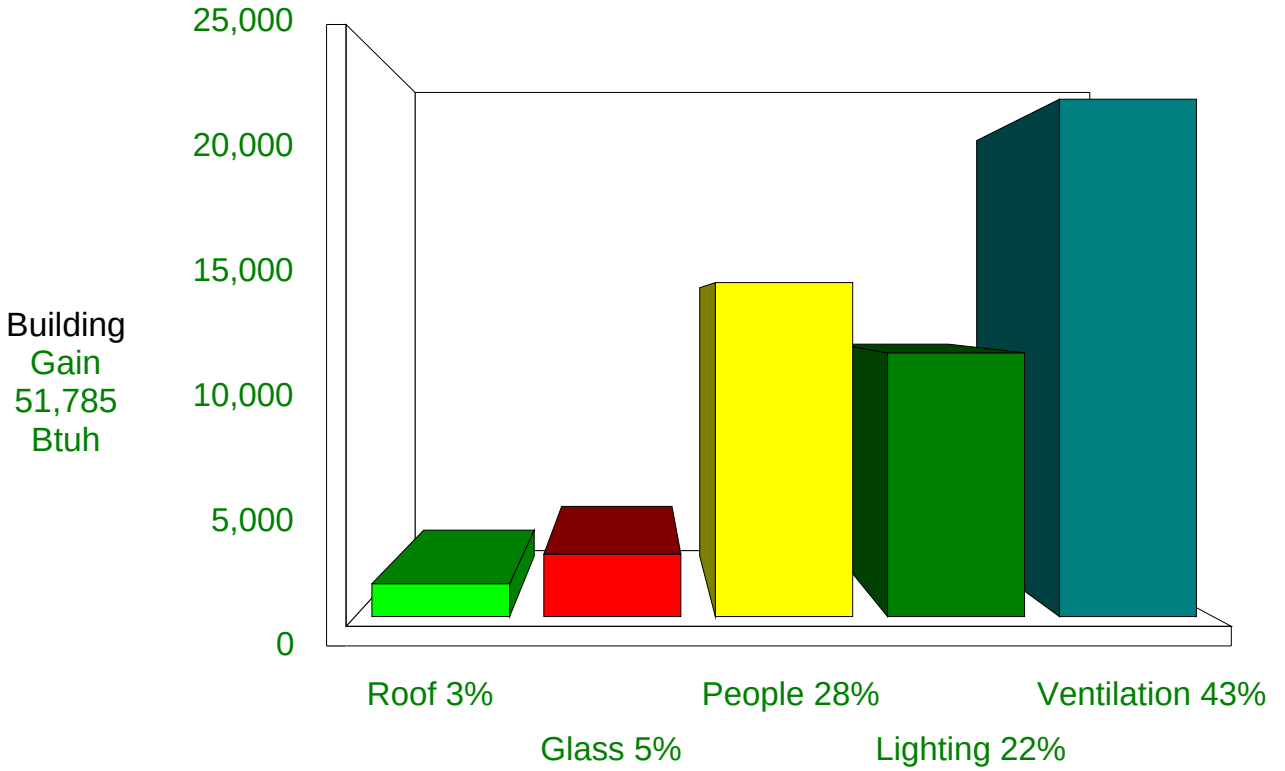


Building
Gain
51,785
Btuh



Building Bar Graph





System 1 System Summary Loads

Component Description	Area Quan	Sen Loss	Lat Gain	Sen Gain	Total Gain
SL: Glazing-Skylight, Double Glaze, Low e Glass, horizontal, u-value 0.55	32	1,056	0	2,658	2,658
Roof: Roof/Ceiling-	1723	1,964	0	1,408	1,408
Basment Floor: Floor-	1728	10,368	0	0	0
Subtotals for structure:		13,388	0	4,066	4,066
People:	27		6,210	8,100	14,310
Equipment:			0	0	0
Lighting:	3305			11,270	11,270
Ductwork:		0	0	0	0
Infiltration: Winter CFM: 0, Summer CFM: 0		0	0	0	0
Ventilation: Winter CFM: 540, Summer CFM: 540		35,030	11,630	10,509	22,139
System 1 System Load Totals:		48,418	17,840	33,945	51,785

Check Figures

Supply CFM:	1,084	CFM Per Square ft.:	4.817
Square ft. of Room Area:	225	Square ft. Per Ton:	38
Volume (ft³) of Cond. Space:	0	Air Turnover Rate (per hour):	0.0

System Loads

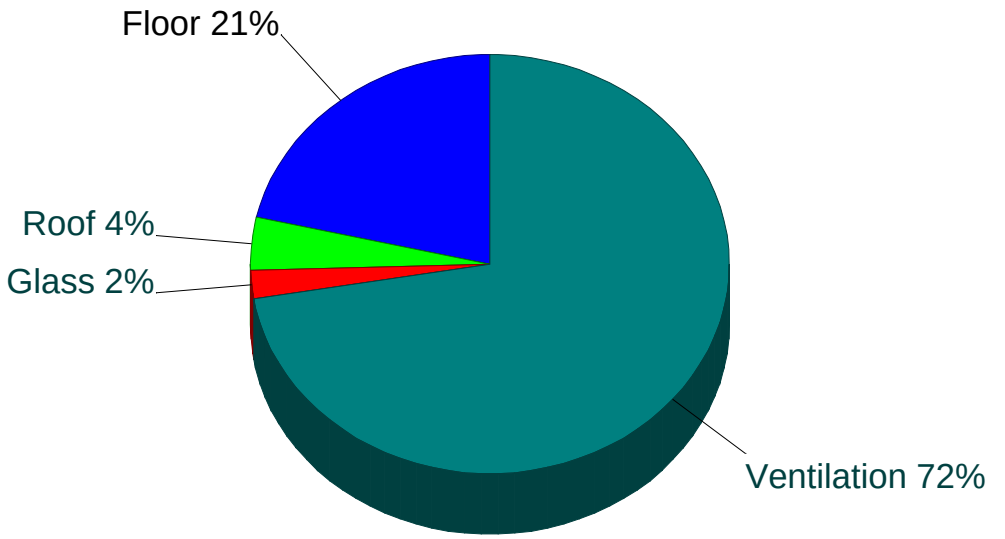
Total Heating Required With Outside Air:	48,418	Btuh	48.418	MBH
Total Sensible Gain:	33,945	Btuh	66	%
Total Latent Gain:	17,840	Btuh	34	%
Total Cooling Required With Outside Air:	51,785	Btuh	4.32	Tons (Based On Sensible + Latent)
			5.95	Tons (Based On 75% Sensible Capacity)

Notes

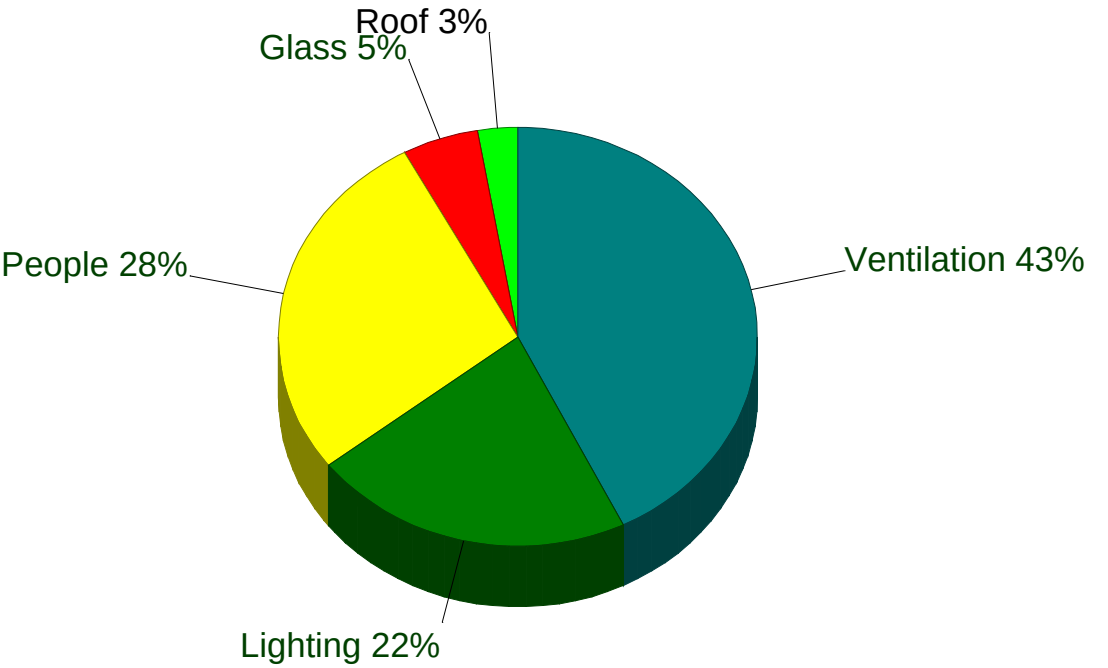
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Be sure to select a unit that meets both sensible and latent loads.

System 1 System Pie Chart

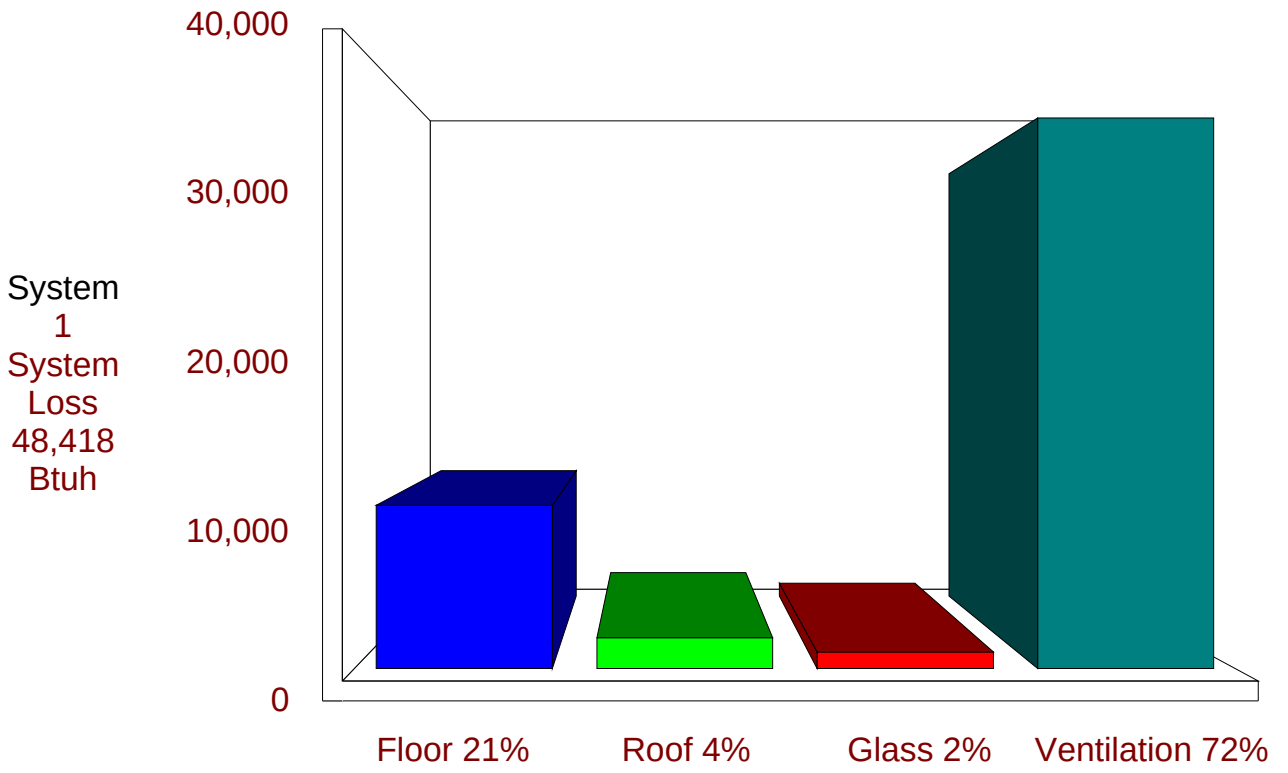
System 1 System Loss 48,418 Btuh

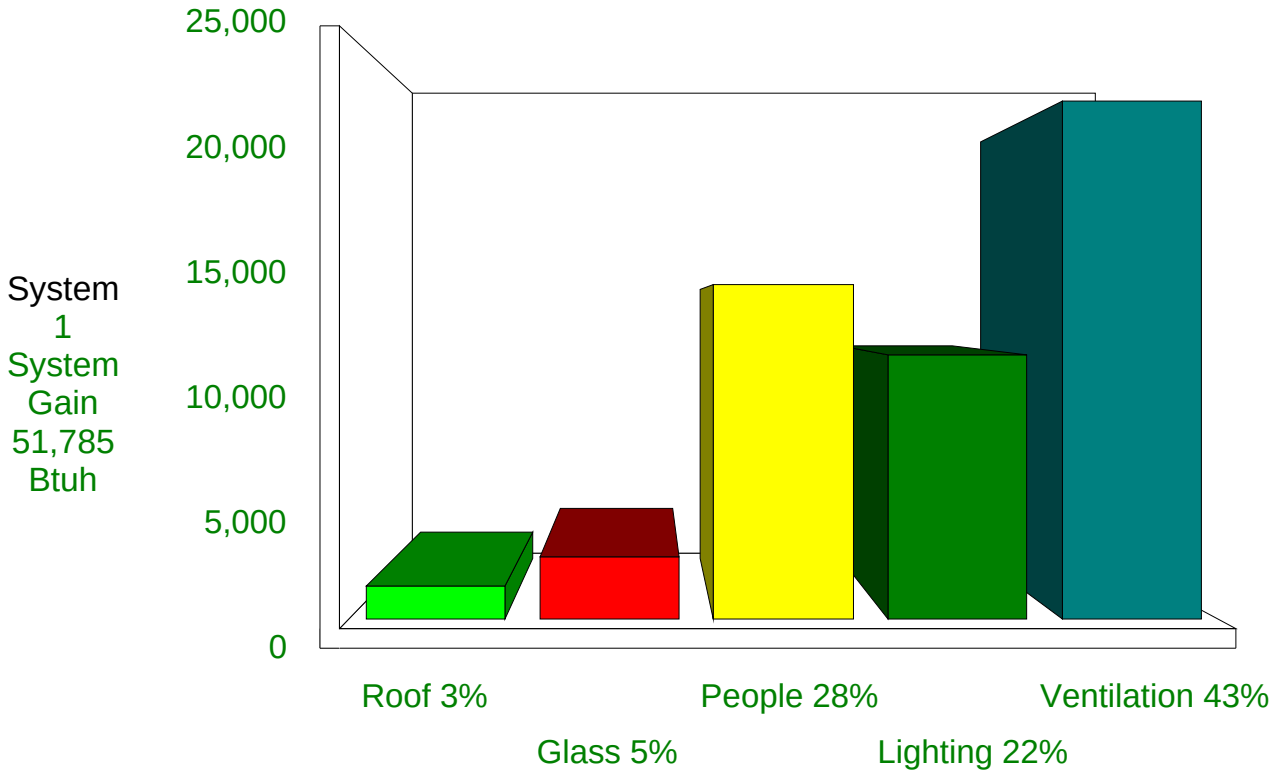


System 1 System Gain 51,785 Btuh



System 1 System Bar Graph





System 1, Zone 1 Summary Loads (Average Load Procedure for Rooms)

Component Description	Area Quan	Sen Loss	Lat Gain	Sen Gain	Total Gain
SL: Glazing-Skylight, Double Glaze, Low e Glass, horizontal, u-value 0.55	32	1,056	0	2,658	2,658
Roof: Roof/Ceiling-	1723	1,964	0	1,408	1,408
Basment Floor: Floor-	1728	10,368	0	0	0
Subtotals for structure:		13,388	0	4,066	4,066
People:	27		6,210	8,100	14,310
Equipment:			0	0	0
Lighting:	3305			11,270	11,270
Ductwork:		0	0	0	0
Infiltration: Winter CFM: 0, Summer CFM: 0		0	0	0	0
System 1, Zone 1 Load Totals:		13,388	6,210	23,436	29,646

Check Figures

Supply CFM:	1,084	CFM Per Square ft.:	4.817
Square ft. of Room Area:	225	Square ft. Per Ton:	63
Volume (ft³) of Cond. Space:	0	Air Turnover Rate (per hour):	0.0

Zone Loads

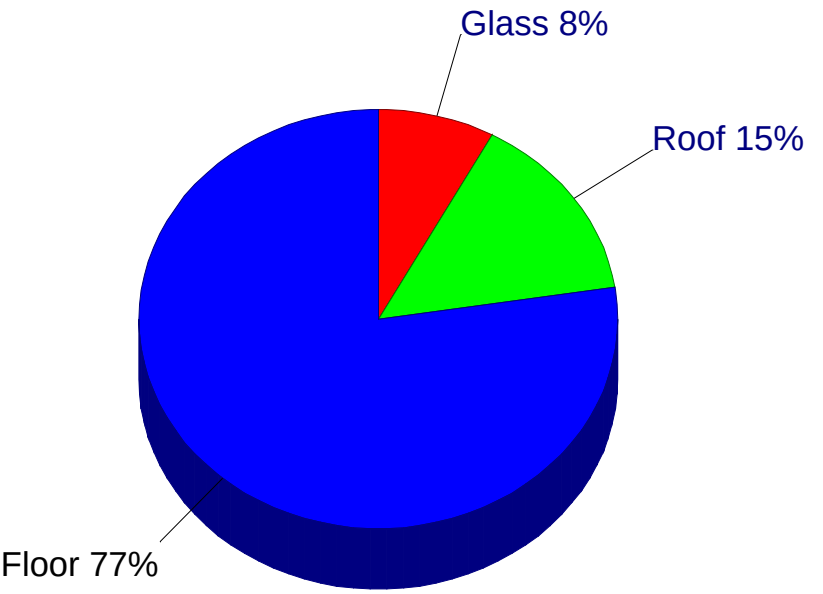
Total Heating Required:	13,388	Btuh	13.388	MBH
Total Sensible Gain:	23,436	Btuh	79	%
Total Latent Gain:	6,210	Btuh	21	%
Total Cooling Required:	29,646	Btuh	2.47	Tons (Based On Sensible + Latent)
			3.60	Tons (Based On 75% Sensible Capacity)

Notes

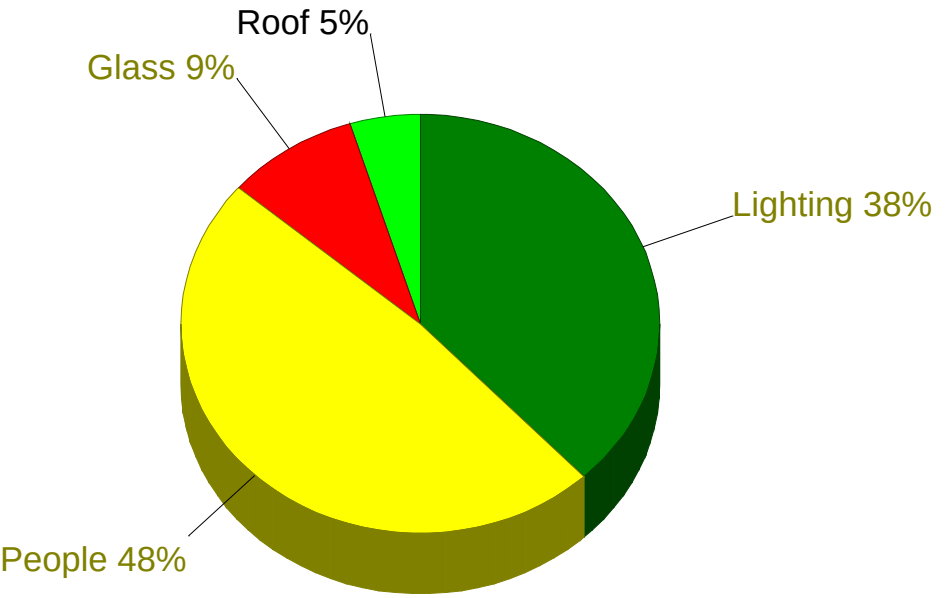
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System 1, Zone 1 Pie Chart

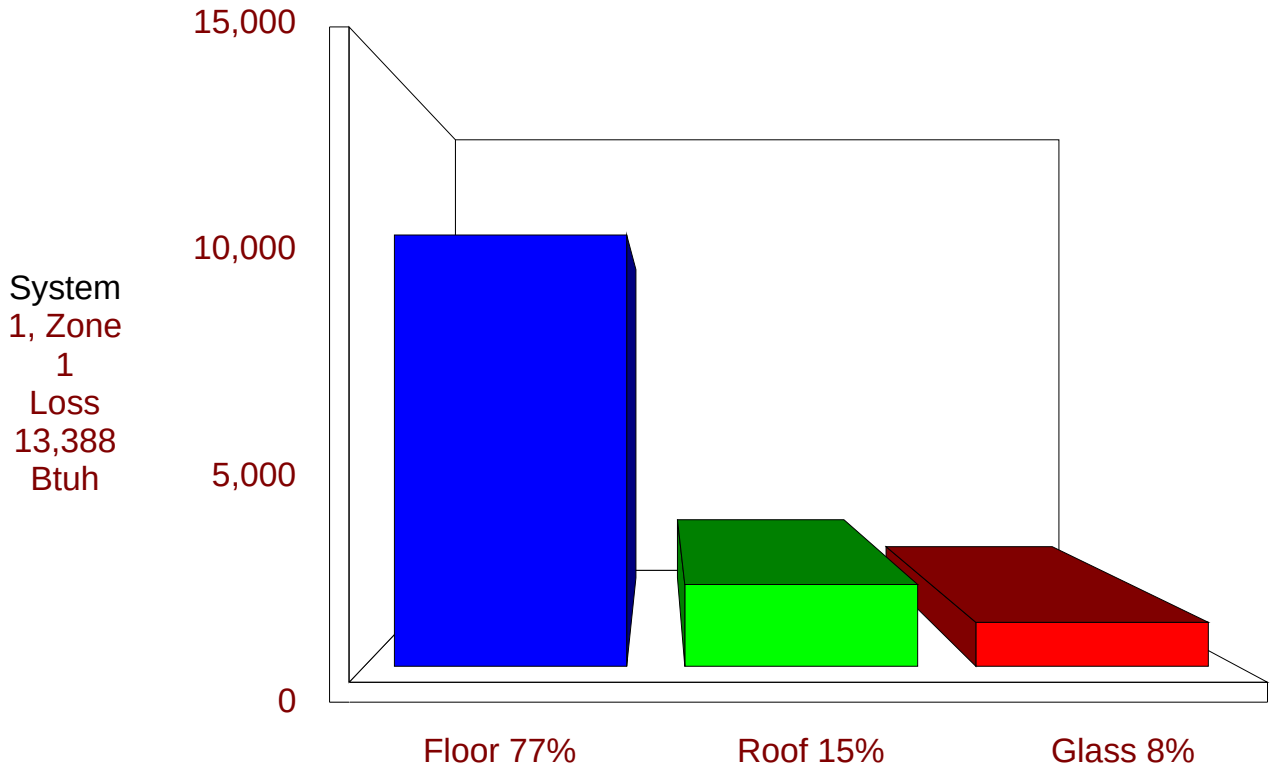
System 1, Zone 1 Loss 13,388 Btuh

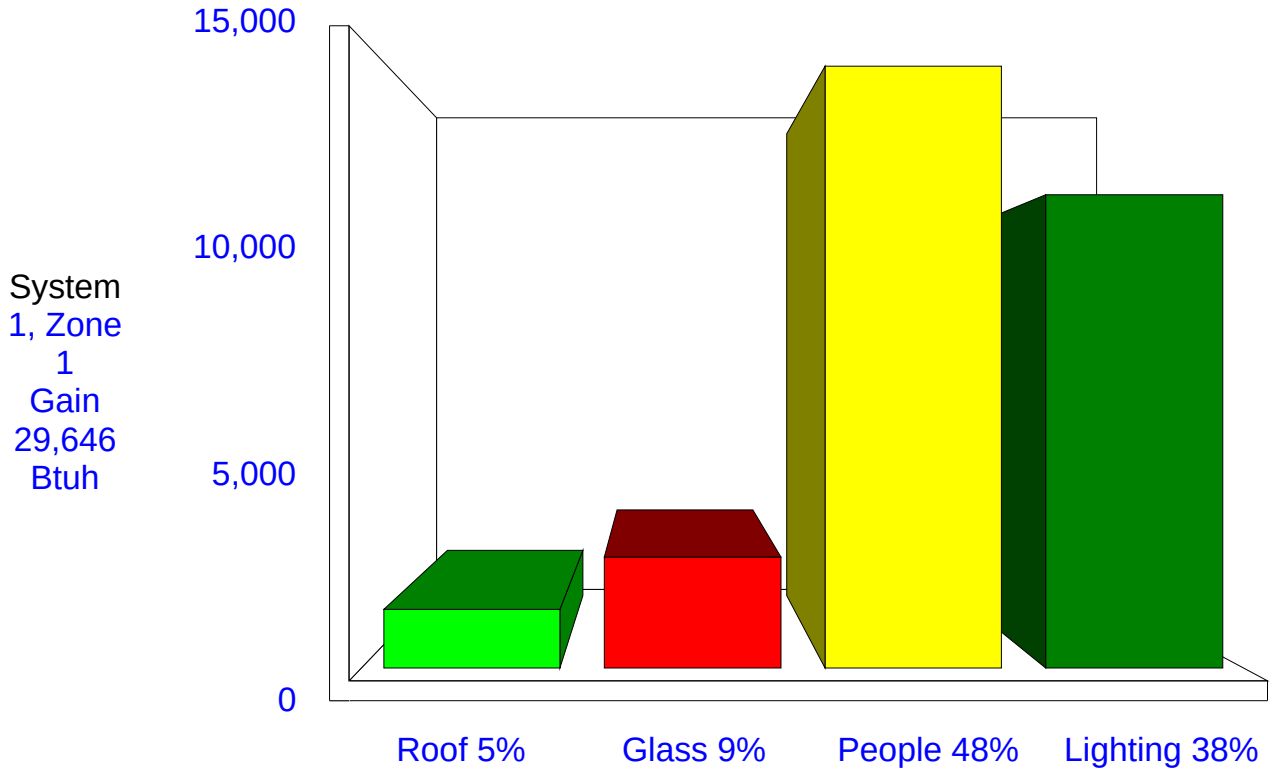


System 1, Zone 1 Gain 29,646 Btuh



System 1, Zone 1 Bar Graph





Detailed Room Loads - Room 1 - Basement (Average Load Procedure)

General

Calculation Mode:	Htg. & clg.		Occurrences:	1	
Room Length:	225.0	ft.	System Number:	1	
Room Width:	1.0	ft.	Zone Number:	1	
Area:	225.0	sq.ft.	Supply Air:	457	CFM
Ceiling Height:	0.0	ft.	Supply Air Changes:	0.0	AC/hr
Volume:	0.0	cu.ft.	Required Vent.:	0	CFM
Number of Registers:	4		Actual Winter Vent.:	418	CFM
Runout Air:	114	CFM	Percent of Supply.:	91	%
Runout Duct Size:	6	in.	Actual Summer Vent.:	228	CFM
Runout Air Velocity:	582	ft./min.	Percent of Supply:	50	%
Design Loss:	0.100	in.wg./100 ft.	Actual Winter Infil.:	0	CFM
Actual Loss:	0.213	in.wg./100 ft.	Actual Summer Infil.:	0	CFM

Item Description	Area Quantity	-U- Value	Htg HTM	Sen Loss	Clg HTM	Lat Gain	Sen Gain
Floor-Basment Floor 1 X 1728	1728	0.100	6.0	10,368	0.0	0	0
Subtotals for Structure:				10,368		0	0
Infil.: Win.: 0.0, Sum.: 0.0	0		0	0	0	0	0
People: 230 lat/per, 300 sen/per:	14					3,220	4,200
Lighting:	1,668						5,688
Room Totals:				10,368		3,220	9,888

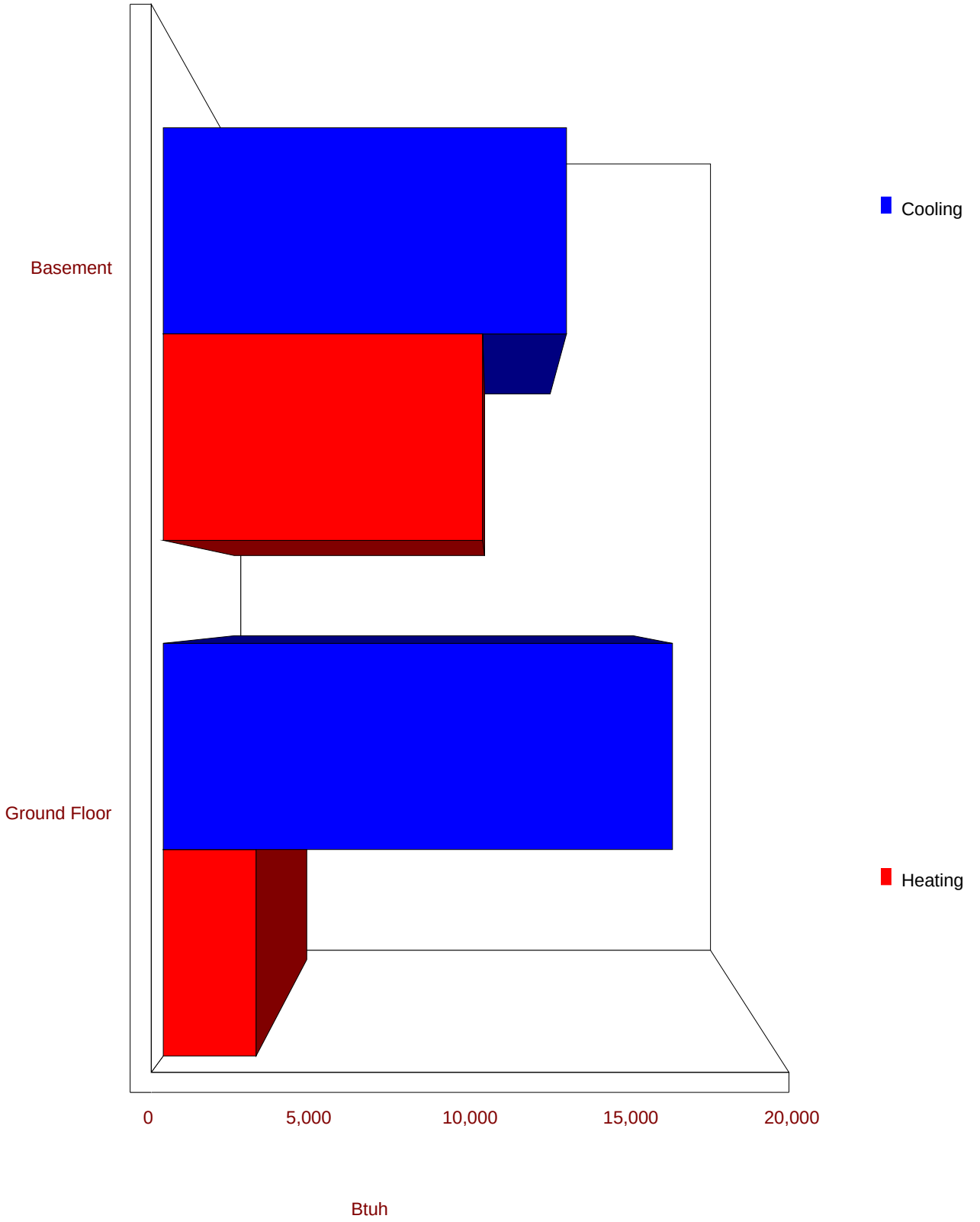
Detailed Room Loads - Room 2 - Ground Floor (Average Load Procedure)

General

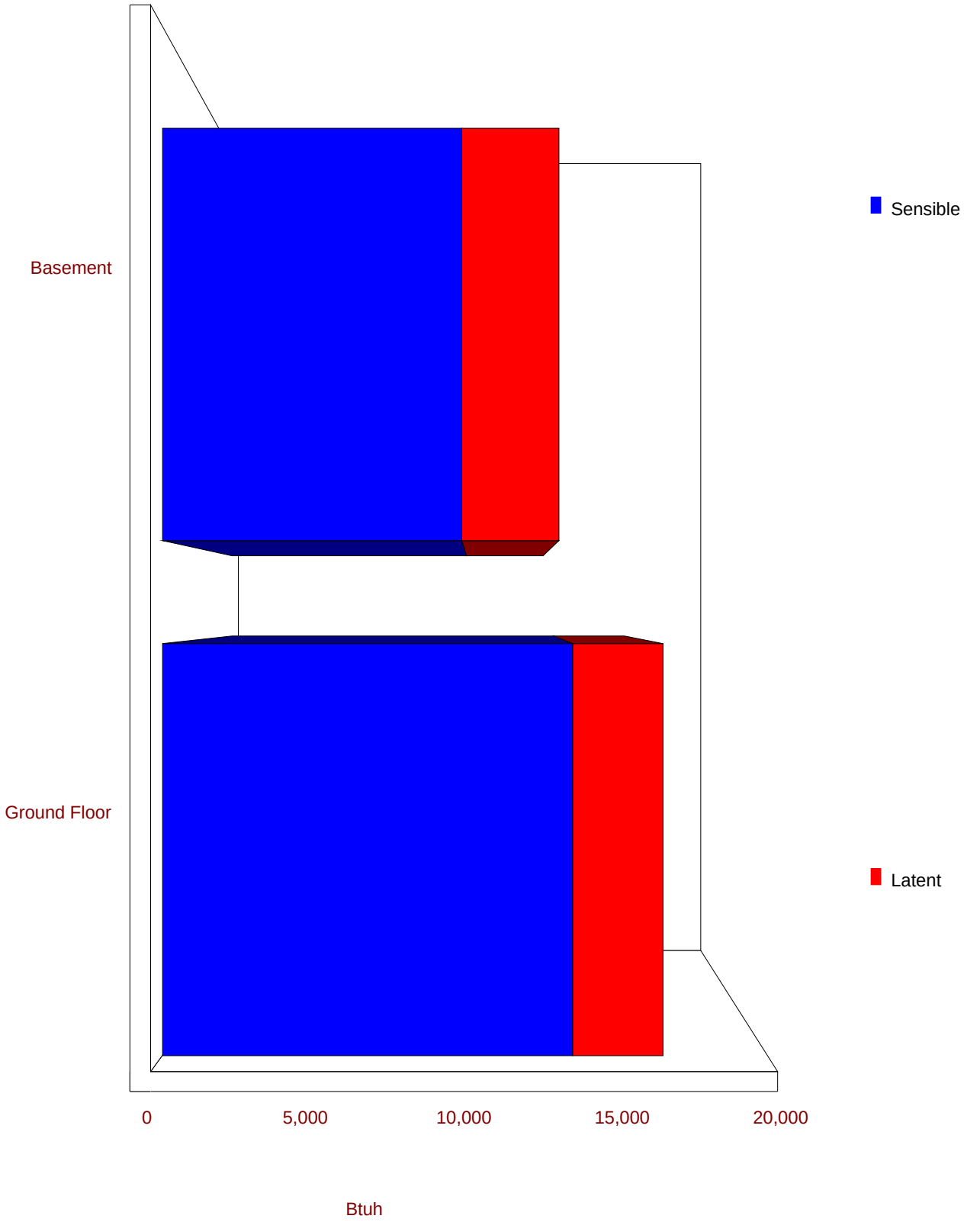
Calculation Mode:	Htg. & clg.		Occurrences:	1	
Room Length:	0.0	ft.	System Number:	1	
Room Width:	0.0	ft.	Zone Number:	1	
Area:	0.0	sq.ft.	Supply Air:	627	CFM
Ceiling Height:	0.0	ft.	Supply Air Changes:	0.0	AC/hr
Volume:	0.0	cu.ft.	Required Vent.:	0	CFM
Number of Registers:	6		Actual Winter Vent.:	122	CFM
Runout Air:	104	CFM	Percent of Supply.:	19	%
Runout Duct Size:	6	in.	Actual Summer Vent.:	312	CFM
Runout Air Velocity:	532	ft./min.	Percent of Supply:	50	%
Design Loss:	0.100	in.wg./100 ft.	Actual Winter Infil.:	0	CFM
Actual Loss:	0.178	in.wg./100 ft.	Actual Summer Infil.:	0	CFM

Item Description	Area Quantity	-U- Value	Htg HTM	Sen Loss	Clg HTM	Lat Gain	Sen Gain
UP-Sky-SL shgc-0.33	32	0.550	33.0	1,056	83.1	0	2,658
UP-Ceil-Roof 45 X 39	1723	0.019	1.1	1,964	0.8	0	1,408
Subtotals for Structure:				3,020		0	4,066
Infil.: Win.: 0.0, Sum.: 0.0	0		0	0	0	0	0
People: 230 lat/per, 300 sen/per:	13					2,990	3,900
Lighting:	1,637						5,582
Room Totals:				3,020		2,990	13,548

Room Cooling and Heating Loads Bar Graphs



Room Cooling Loads Bar Graphs



System 1 Room Load Summary

No	Room Name	Area SF	Htg Sens Btuh	Htg Nom CFM	Run Duct Size	Run Duct Vel	Clg Sens Btuh	Clg Lat Btuh	Clg Nom CFM	Air Sys CFM
---Zone 1---										
1	Basement	225	10,368	137	4-6	582	9,888	3,220	457	457
2	Ground Floor	0	3,020	40	6-6	532	13,548	2,990	627	627
	Ventilation		35,030				10,509	11,630		
	System 1 total	225	48,418	177			33,945	17,840	1,084	1,084

System 1 Main Trunk Size:	12x16	in.
Velocity:	871	ft./min
Loss per 100 ft.:	0.098	in.wg

Cooling System Summary

	Cooling Tons	Sensible/Latent Split	Sensible Btuh	Latent Btuh	Total Btuh
Net Required:	4.32	66% / 34%	33,945	17,840	51,785
Recommended:	5.95	75% / 25%	53,521	17,840	71,361